

Session Objectives

1. Student will be able to understand the principles and concepts of osteopathy the cranial field including the components of the primary respiratory mechanism (PRM).
2. Student will be able to understand the developmental anatomy and physiologic and non-physiologic motions that occur at the sphenobasilar synchondrosis (SBS).
3. Student will be able to understand the mechanisms and movement of the sphenoid and occiput that create cranial strain patterns and to be able to identify those patterns through a cranial vault hold.
4. Student will be able to understand cranial nerves function and identify potential presentations if there is nerve compression and the associated cranial bones that might be involved.
5. Student will understand the application and indications that osteopathy in the cranial field can be utilized for in different patient conditions and presentations.

Case #1

- A 2-month-old baby girl is brought to the office with her mom complaining of her neck being twisted and a misshaped head.
- Mom reports that since delivery the baby does not like to turn her head to the right.
- She heard that osteopathic manipulative medicine can help her daughter and she has more questions for you before starting treatment.

Case #1

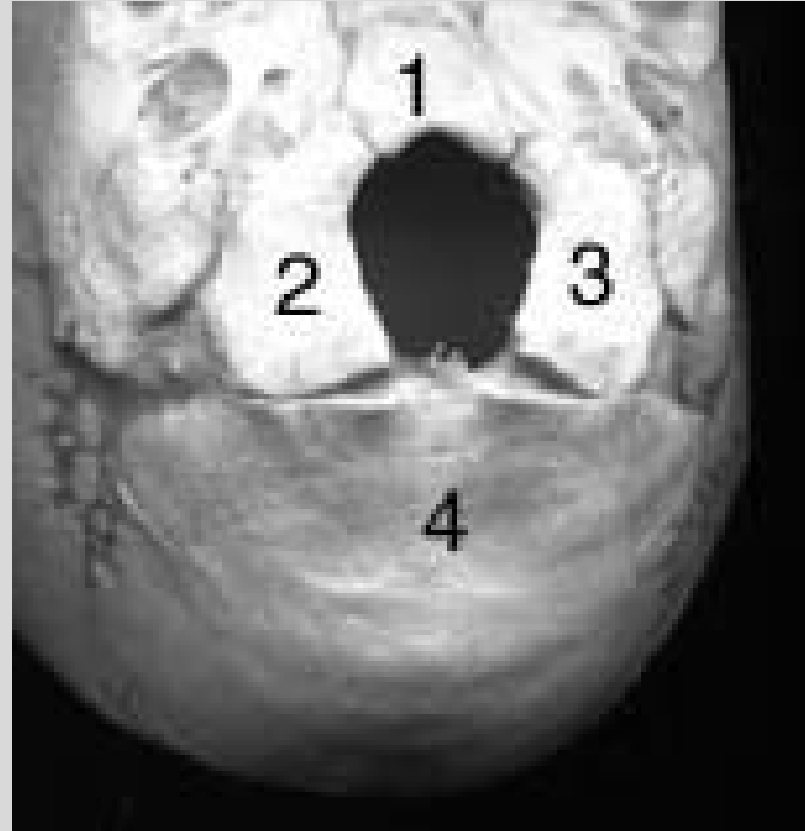
- How would you describe osteopathic manipulative medicine in the cranial field or Osteopathic Cranial Manipulative Medicine (OCMM)?
- How do you approach the patient?
- How do you explain to the mom through anatomy and physiology how osteopathy in the cranial field can help her baby?

Application of Cranial Osteopathic Manipulative Medicine formally known as Osteopathy in the Cranial Field (OCF)

- History – Trauma
- Pediatrics – Delivery and Newborn considerations
- Observation
- Palpation
- Treatment

Cranial Osteology - infant

- At birth:
- Occiput in 4 parts
- Sphenoid in 3 parts
- Temporal in 3 parts
- Maxillae in 2 parts
- 6 Fontanelles
- No mastoid process

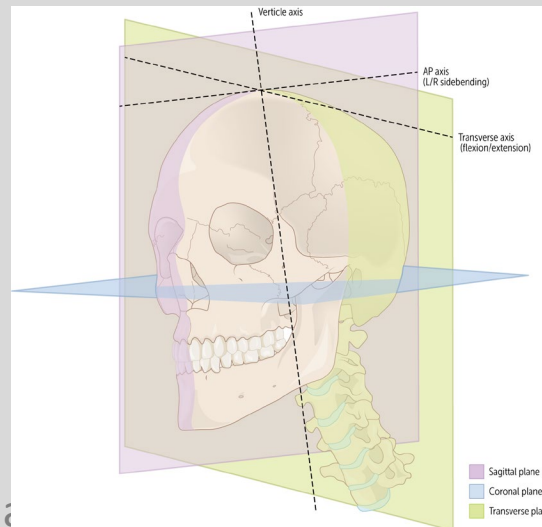
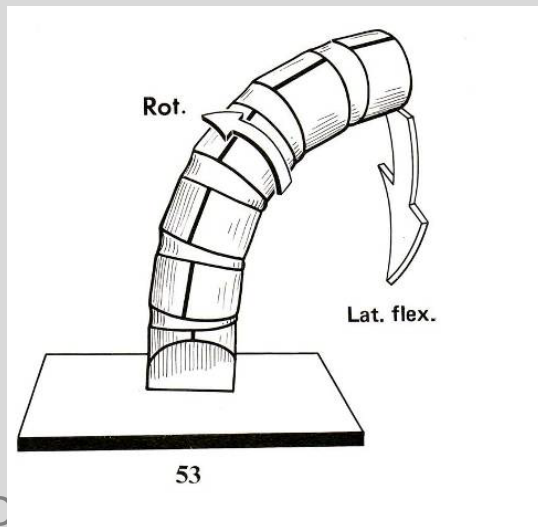
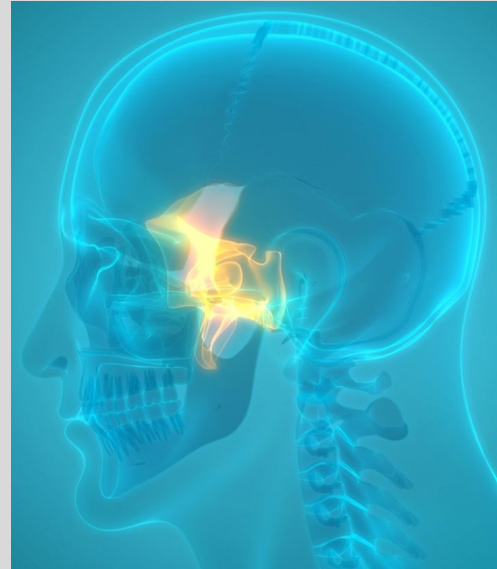
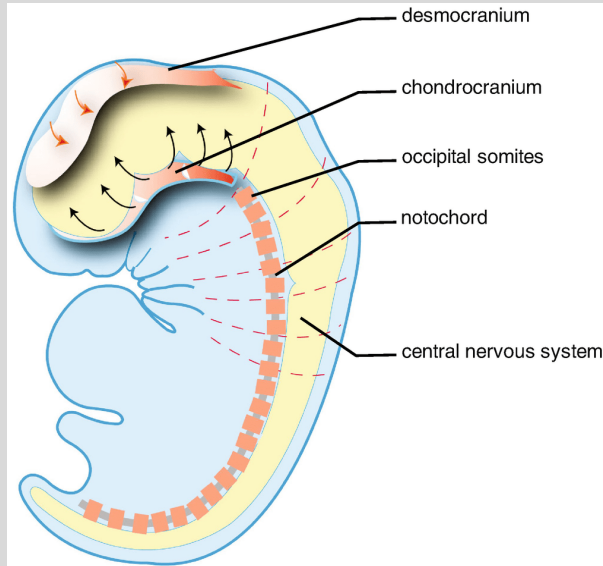




Primary Respiratory Mechanism

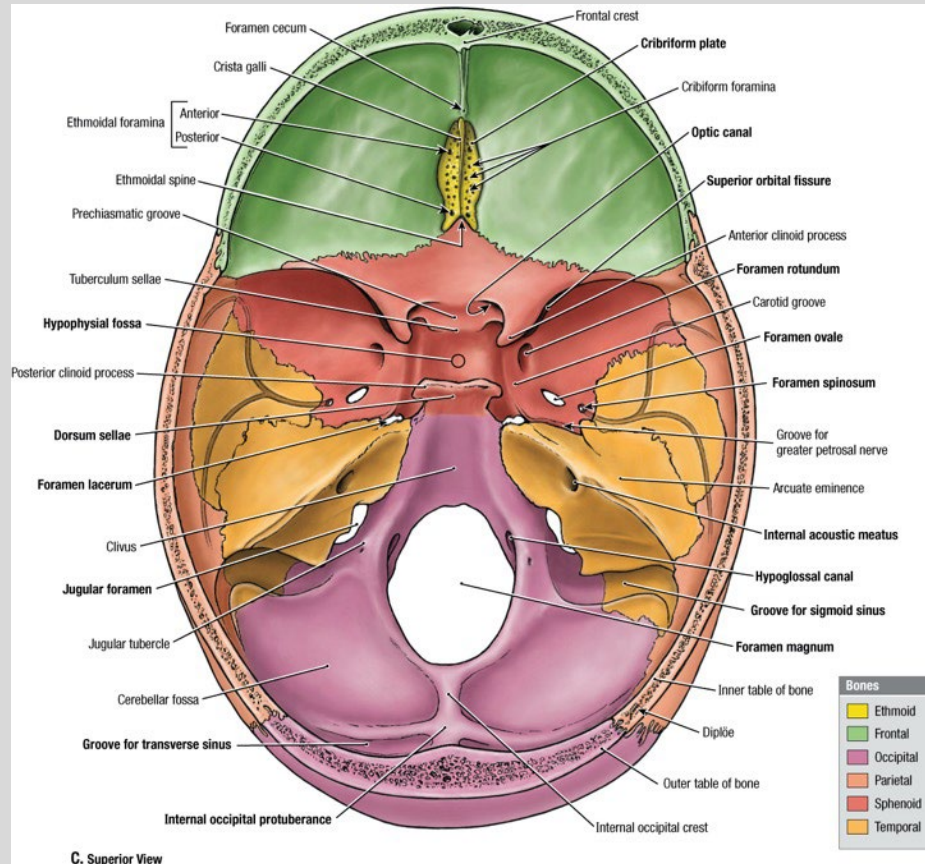
1. Inherent *motility* of the brain and spinal cord.
2. Fluctuation of the CSF.
3. Movement of the intracranial and intraspinal membranes.
4. Articular *mobility* of the cranial bones.
5. Involuntary *mobility* of the sacrum between the ilia.

Development of the SBS



- The SBS is the articulation between the sphenoid and occiput.
- Forms from cartilage dubbed "cranial vertebra"
- Continuation of the spine with a change in orientation
- Similar motion can occur at the SBS as in the spine (flexion/ extension, rotation and side bending)

Strain patterns – etiology



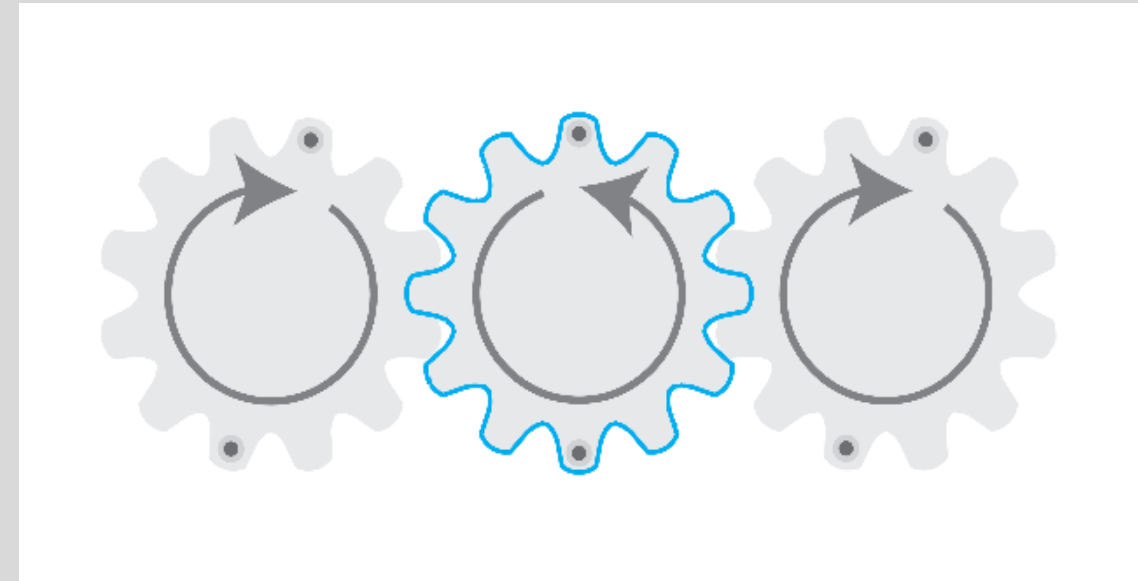
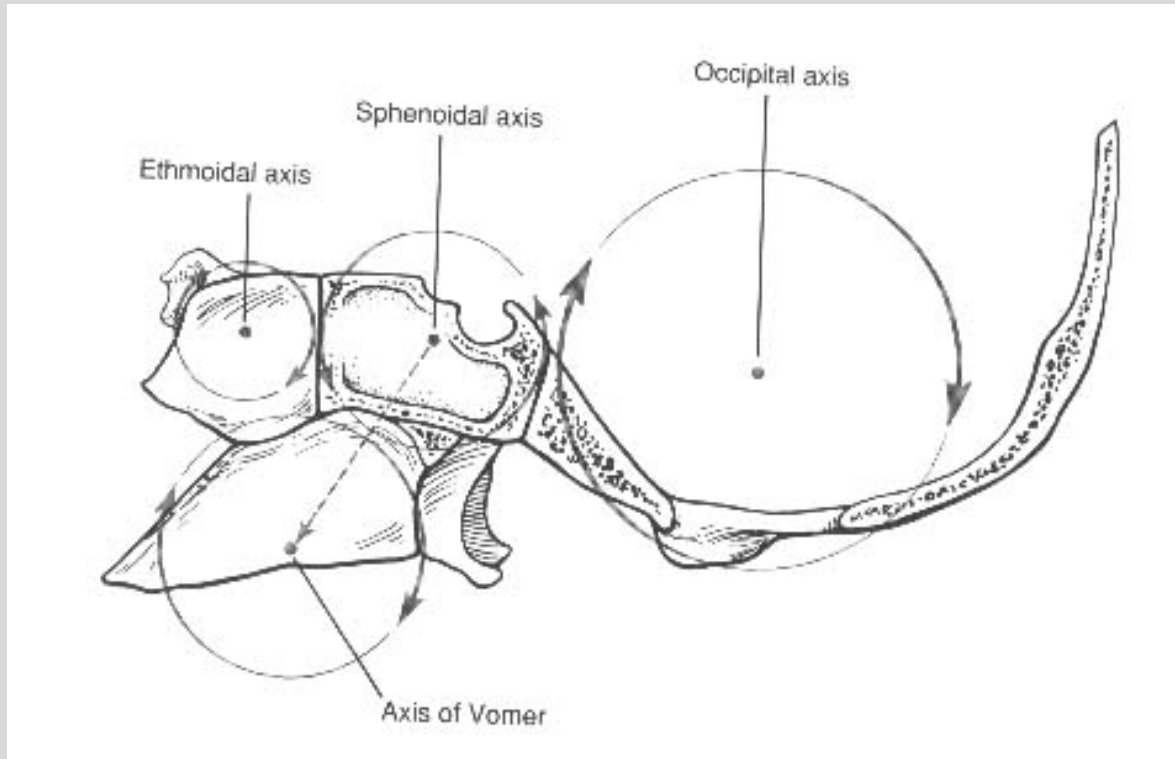
- Traumatic forces produce strain within the base, focused at the SBS and surrounding area, and this will produce an abnormal motion pattern.
- Sutherland through observations on himself and patients.
- Describes these cranial patterns as a schematic to allow for systematic evaluation and documentation
- Most patients will have a predominate pattern due to birth or subsequent patterns

Grant's Atlas of Anatomy, 14e, 2017

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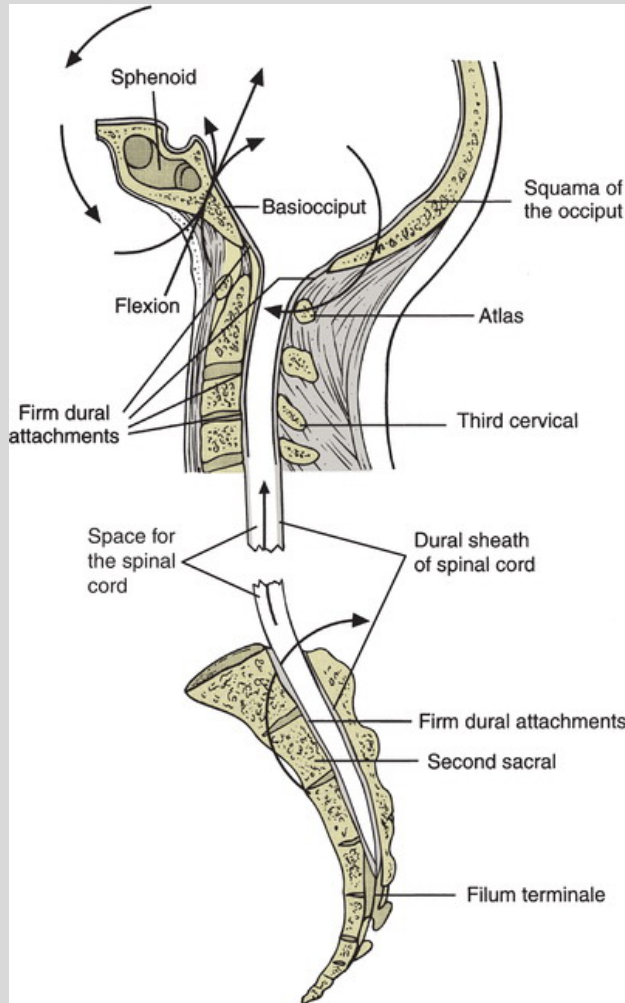
Movement of the Cranial Base



- The midline bones move in a gearlike manner around transverse axes. This motion is called flexion (inhalation phase) and extension (exhalation phase)



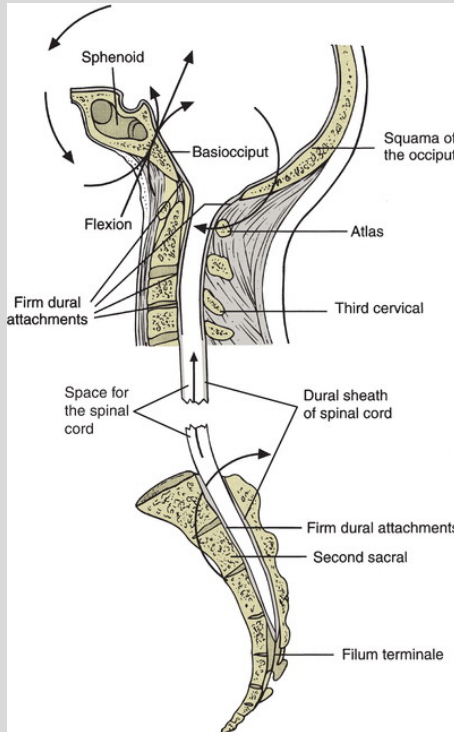
Key dural connections - Core link



- The cranial dura continues down the spinal canal and attaches to C2&C3 and the sacrum
- With craniosacral flexion the sacrum counternutates (or rotates posteriorly)
- With craniosacral extension the sacrum nutates (rotates anteriorly)

From: **38 Osteopathic Cranial Manipulative Medicine**

Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research, 4e, 2018



Legend:

Craniosacral mechanism with arrows indicating direction of motion during flexion phase of physiologic motion.

The motion pattern of the cranium may also be altered into a strain pattern by a variety of other factors.

- Fascial pull from below
- Local cranial dysfunction
- Spinal and/or pelvic asymmetry or dysfunction

Case #1

- After a thorough explanation the mom agrees to treatment with OCMM.
- You place your hands on the head for the vault hold. The mom ask you what are you feeling for?

Characteristics of Strain Patterns

- Axis (or Axes) of Rotation - where are they located and orientation
- Motion - Direction of Rotation of the Sphenoid and Occiput on the axes
- Nomenclature - Generally named for the direction of motion of the basisphenoid relative to the basiocciput
- Morphology - General shape of cranium and regional characteristics
- Etiology – describe potential source for dysfunction
- Palpatory Diagnosis – to be able to described exaggerated motions through vault contact



Cranial strain patterns at the SBS

Physiologic – generally adaptive and generally do not arrest flexion & extension

- Flexion (Inhalation)
- Extension (Exhalation)
- Torsion
- Sidebending and rotation

Non-Physiologic – generally maladaptive, restrict motion and disrupts the midline at the SBS

- Vertical Shears
- Lateral Shears
- Compression

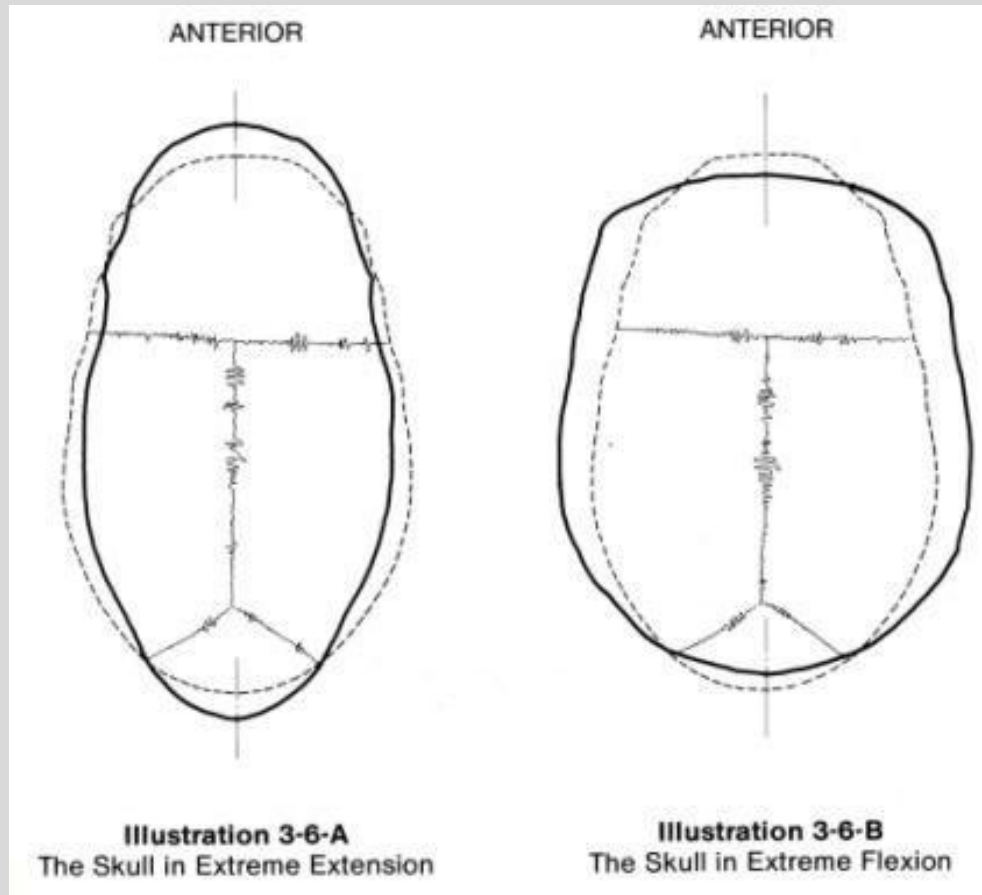
Flexion / Extension Strain Patterns

- Flexion and extension occurs about **two parallel transverse axes**.
 - Sphenoid – through the body of the sphenoid at the level of the SS pivot
 - Occiput - above jugular processes through foramen magnum
- The sphenoid and occiput rotate in **opposite** directions on their axes.
- Physiologic motion





Flexion or Extension Cranial Strain Patterns



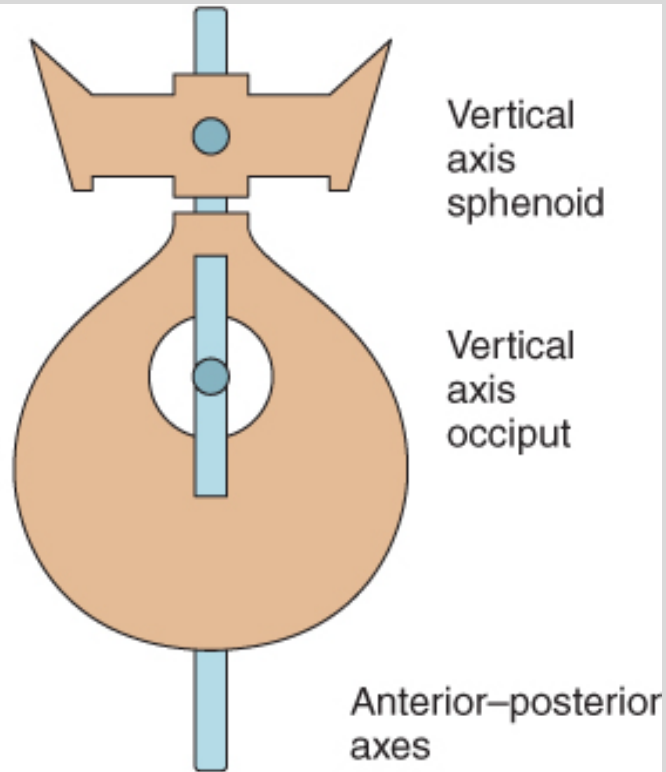
- Like the spine, if either flexion or extension becomes exaggerated, with motion in the opposite direction restricted, this is considered dysfunction.

Case #1 continued

- As you are palpating the head you feel the fingers of your right hand spread out more and are being pulled down to the right inferiorly towards the feet.
- Which strain pattern are you feeling?
- What may be contributing to the strain pattern?
- What visual findings would you expect with this pattern?

Sidebending/Rotation

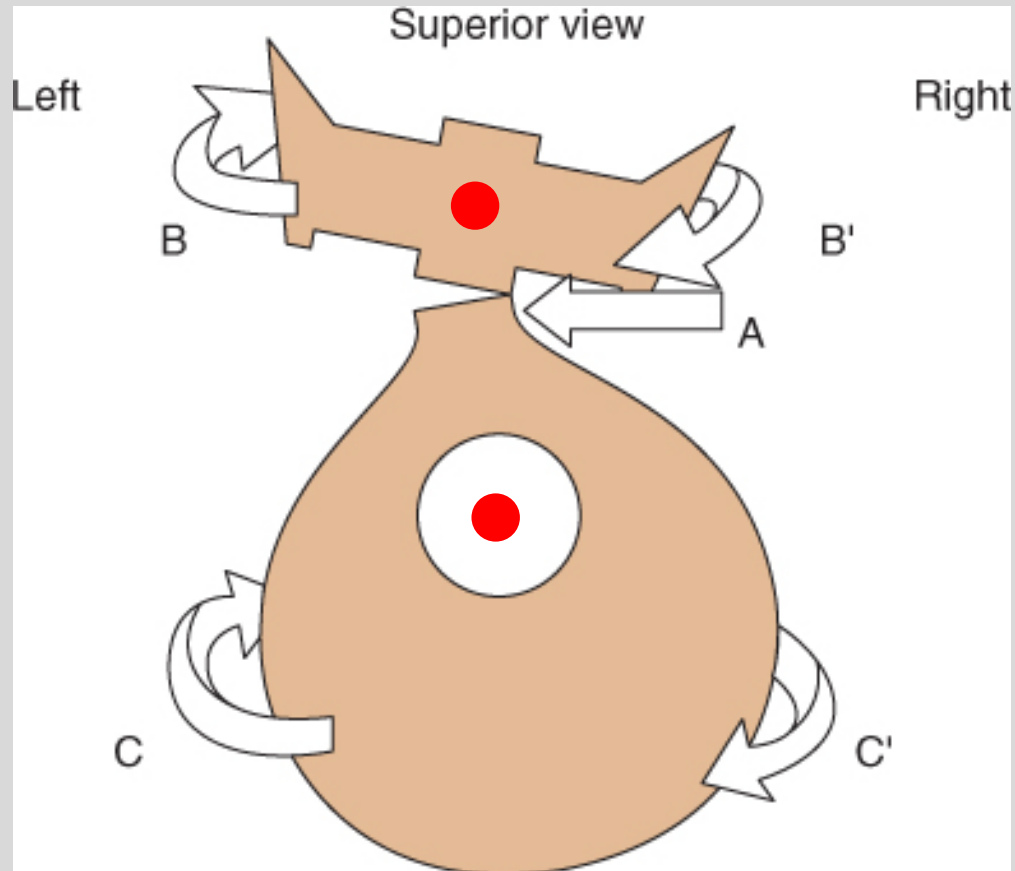
Superior view



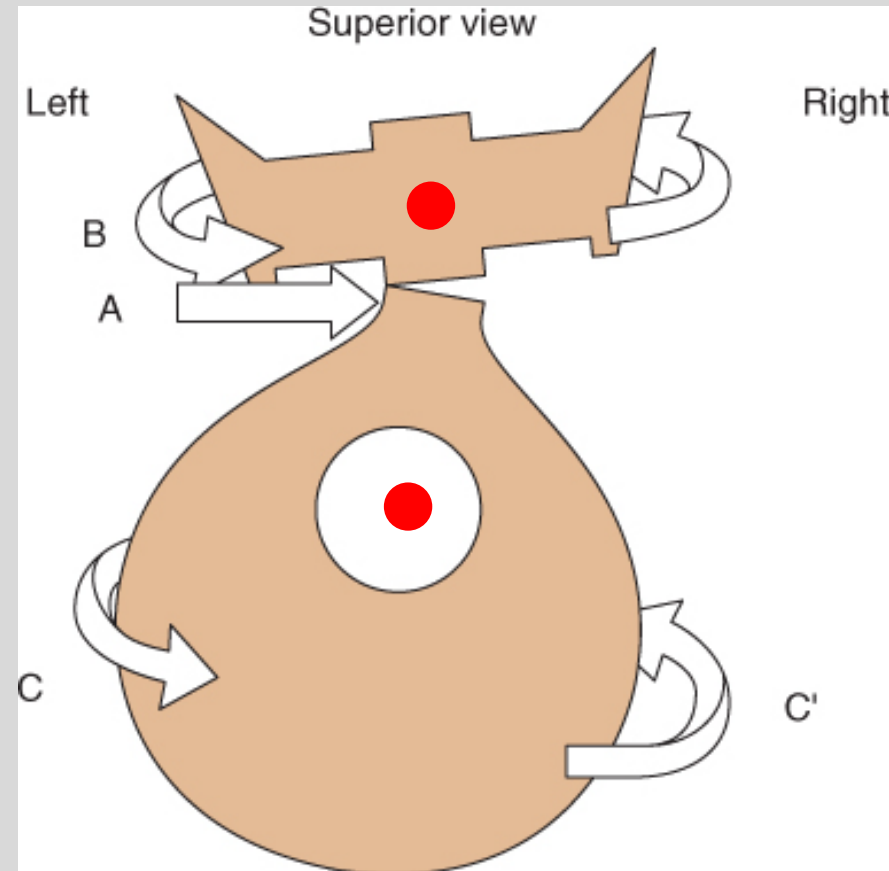
- 3 Axes in total
- Sidebending occurs when the sphenoid and occiput move in opposite directions around 2 parallel vertical axes, one through the foramen magnum of occiput and the other through the body of sphenoid.
- Rotation occurs as both sphenoid and occiput rotate in the same direction around an A-P axis.
- They rotate to the side of the convexity which bringing that side more inferior.

Rotation about 2 vertical axes of the occiput and sphenoid in OPPOSITE direction ->sidebending

- Left SB-Rot

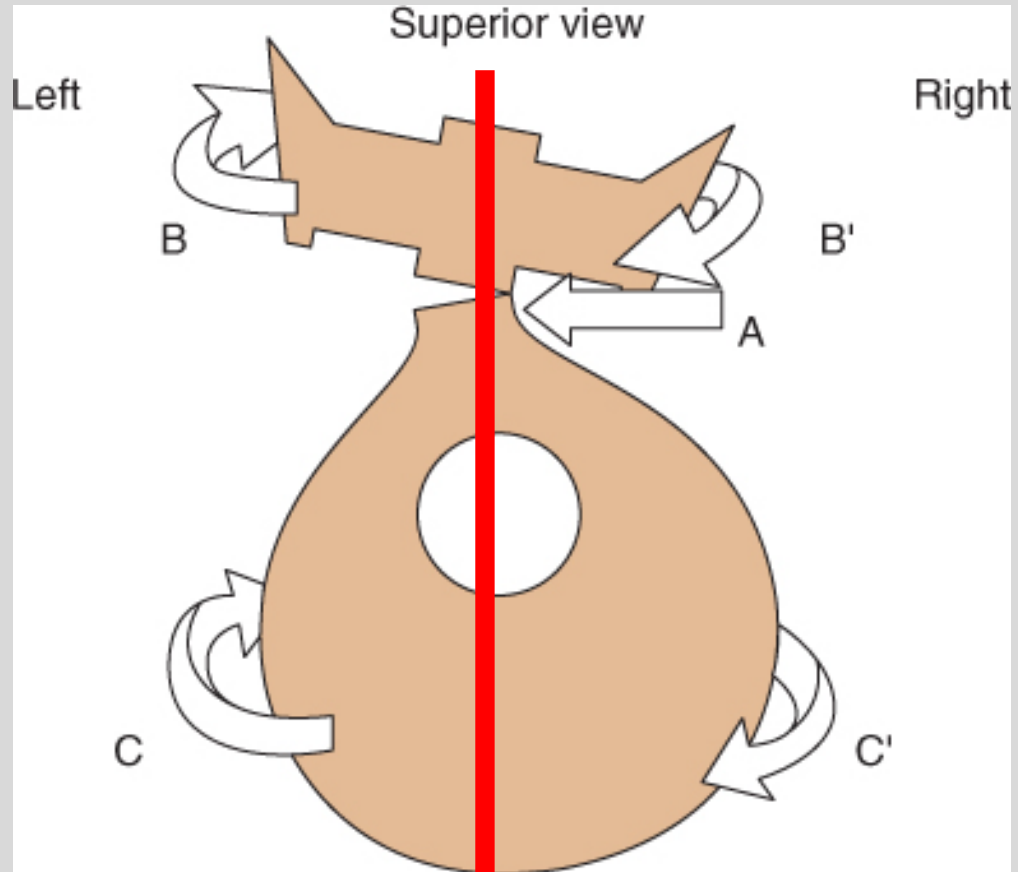


- Right SB-Rot

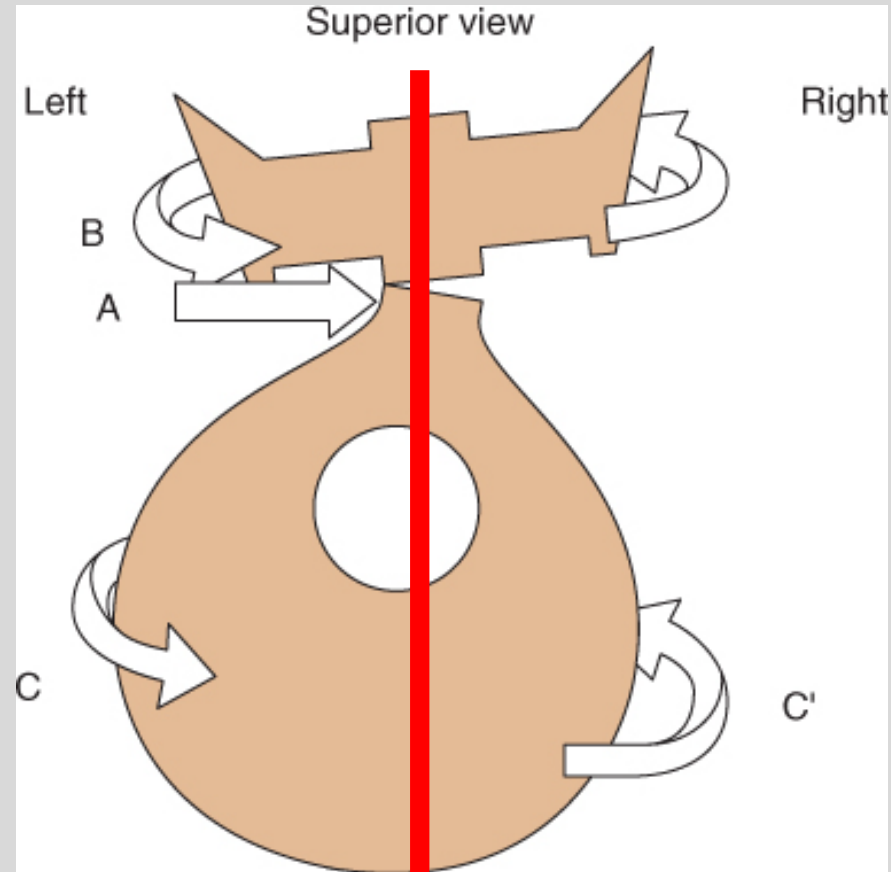


Rotations of the occiput and sphenoid on an AP Axis in the SAME direction

- Left SB-Rot



- Right SB-Rot



Case #1 Continued

- You notice that the baby has decreased rotation to the right. What muscle can be involved with the case?
- What cranial nerve is associated with it?
- Where does that cranial nerve exit the skull?
- What other cranial nerves exits with that nerve and what bodily functions can be affected by irritation of those nerves?

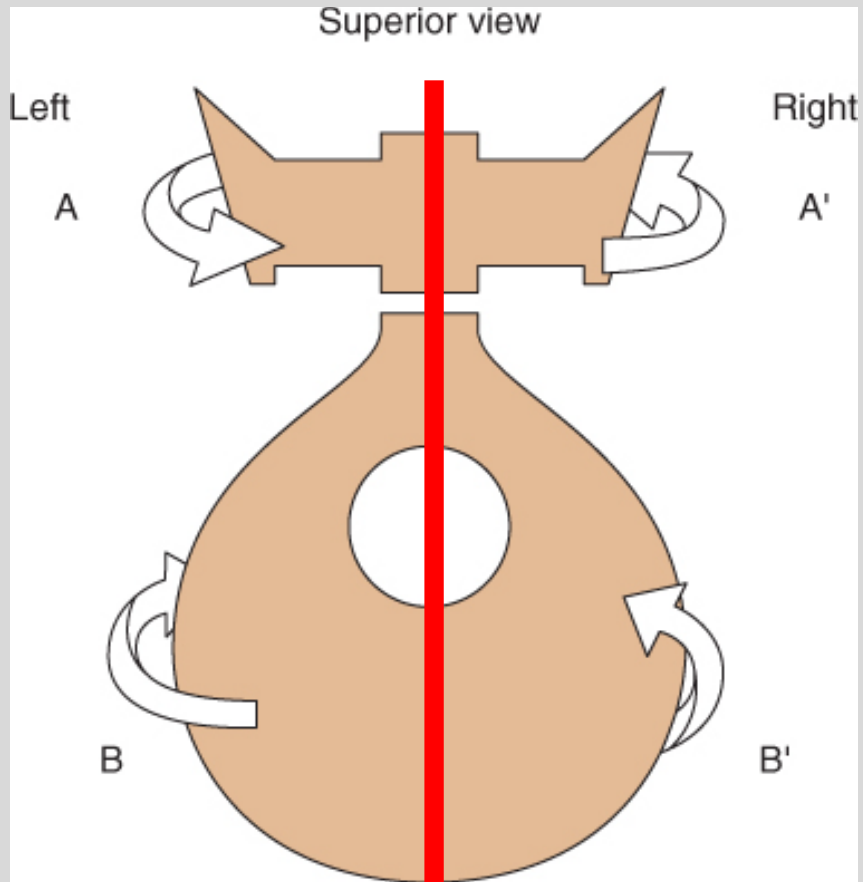
Case #1 continued

- You were able to help with the torticollis after one visit but the baby's head is still misshaped. Looking picture below which strain pattern is still evident?
- What is the medical terminology for a misshaped head?
- How can OMT help with that diagnosis?

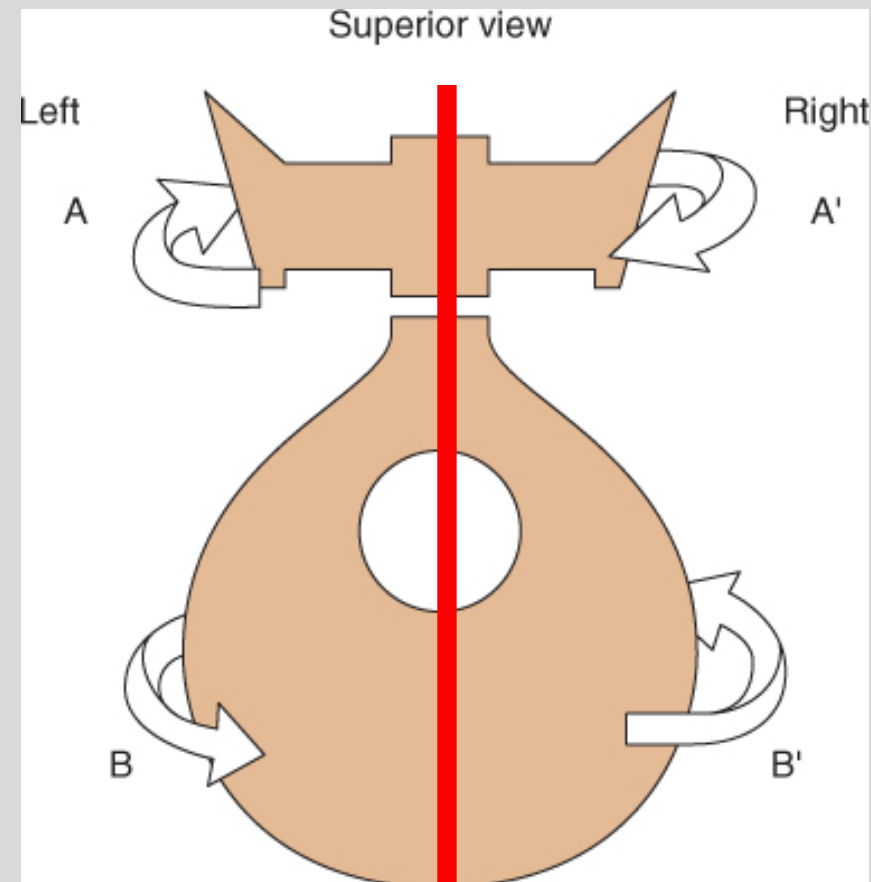


Torsion Strain Pattern

Left Torsion



Right Torsion





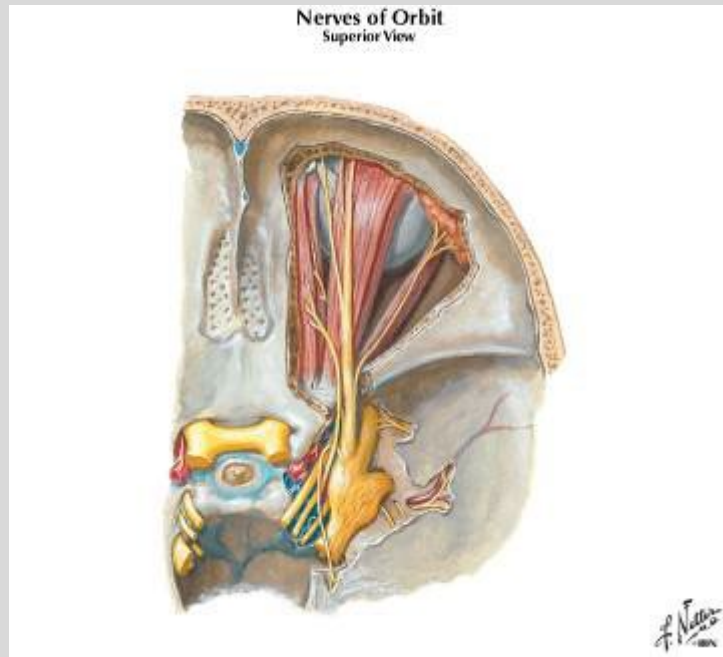
- American Academy of Pediatrics (AAP) released its recommendation in 1992 that infants be placed down for sleep in a nonprone position.
- Helped to decrease rate of Sudden Infant Death Syndrome (SIDS) > 50%
- Increased incidence of plagiocephaly



Extra-ocular muscle disorders

Strabismus / Nystagmus

CN 3,4,6 pass through the superior orbital fissure of the sphenoid.



- CN 3 - oculomotor nerve - eye's movements & constriction of the pupil
- CN 4 - trochlear nerve - superior oblique muscle
- CN 6 - abducens Nerve - lateral rectus muscle

Duane's Retraction Syndrome (DRS)

- Congenital absence or poor development of the abducens nuclei, with aberrant innervation by the oculomotor nerve, that leads to impaired ocular motility.
- 80% of these cases are unilateral and patients have limited abduction, adduction or both.
- Amblyopia occurs in 3 to 25% of patients due to strabismus rather than a refractive difference

Duane's Retraction Syndrome

- While DRS is characterized by a miswiring of the lateral and medial recti muscles, a similar clinical picture may result from various mechanical, anatomical, and innervational disorders.
- The abducens nerve, responsible for abduction of the eye.
- Dural attachments cause the nerve to angulate as it passes under the petrosphenoidal ligament (Gruber's ligament).
- Due to this anatomic relationship, compression or stretching of the nerve may be the cause of various cranial nerve palsies.
- Any change or interruption to the wiring of the lateral rectus may compromise its function.
- Treatment for patients with DRS may include corrective lenses or eye muscle surgery to improve head position but it does not normalize horizontal motility.



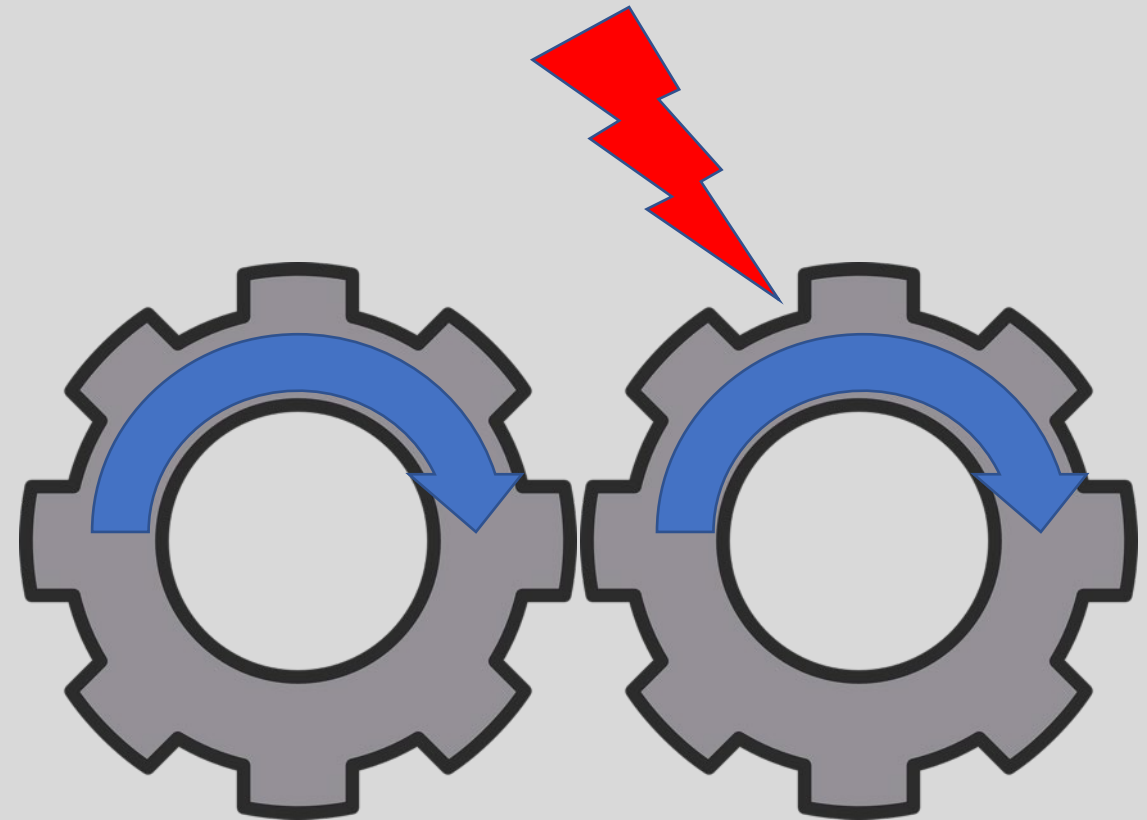
SBS Cranial Strain patterns

Physiologic

- Flexion (Inhalation)
- Extension (Exhalation)
- Torsion
- Sidebending and rotation

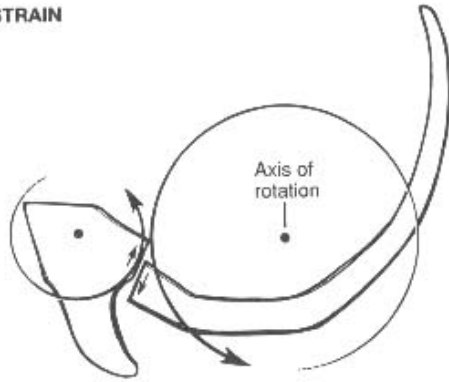
Non-Physiologic

- Vertical Shears
- Lateral Shears
- Compression

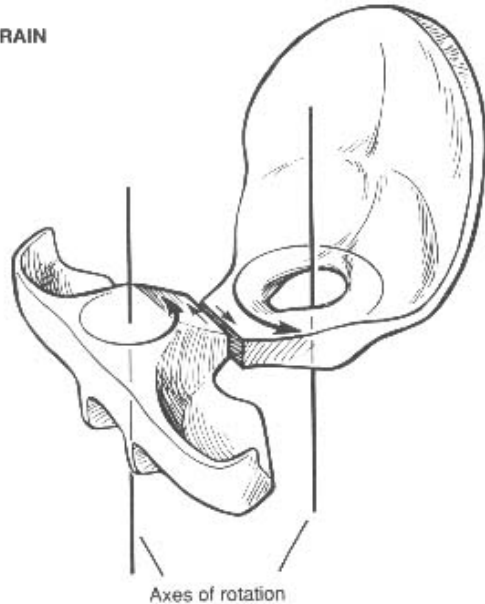


Non-Physiologic Strain Patterns

VERTICAL STRAIN



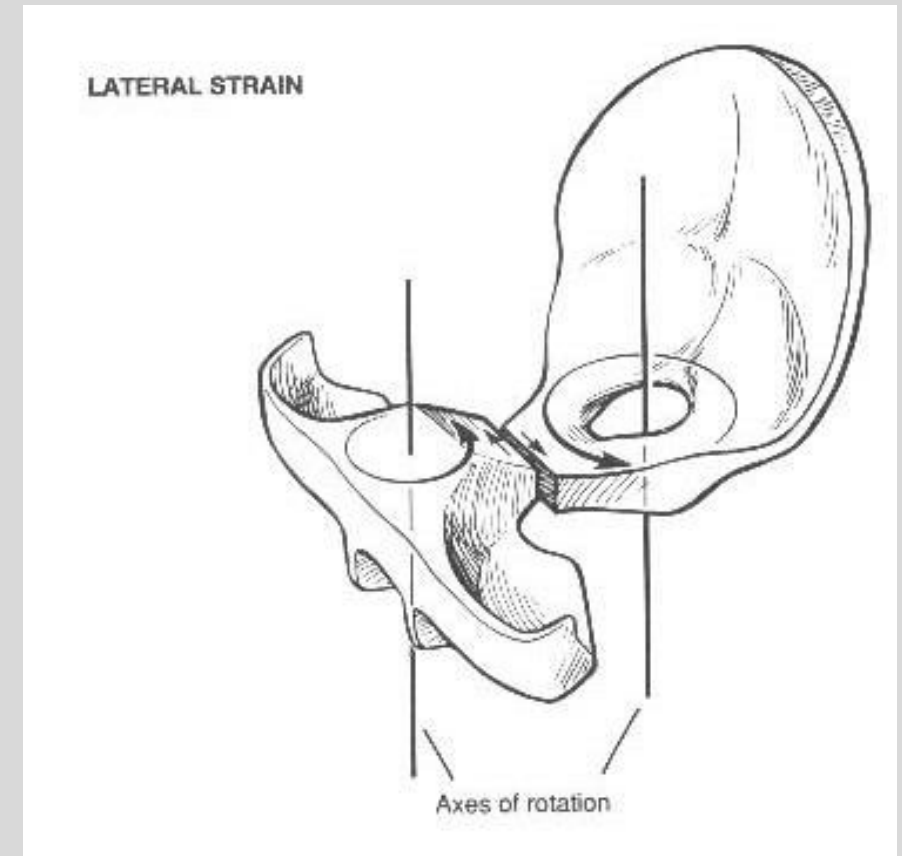
LATERAL STRAIN



- Vertical and lateral strains are non-physiologic, as they involve a shear at the SBS.
- Usually due to trauma that directly affects the SBS disrupting the midline.
- Typically symptomatic and interfered with normal motion thus having a pathologic effect.

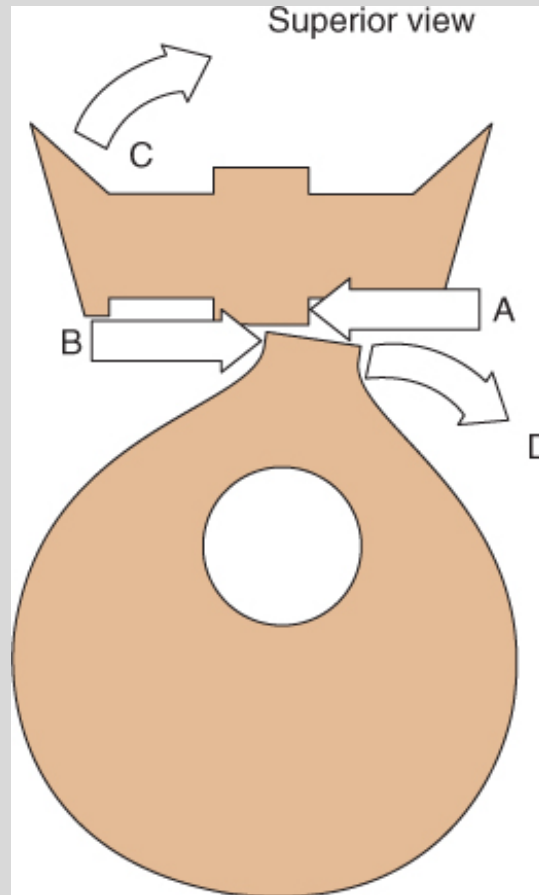
Lateral Strains: “Parallelogram Head”

- Two parallel vertical axes
- The sphenoid and occiput rotate in the **same** direction.
- Lateral strains are named for the direction the basisphenoid moves towards.
- Non-physiologic strain may have varying and complex causes resulting in findings not exactly fitting the expected parallelogram.

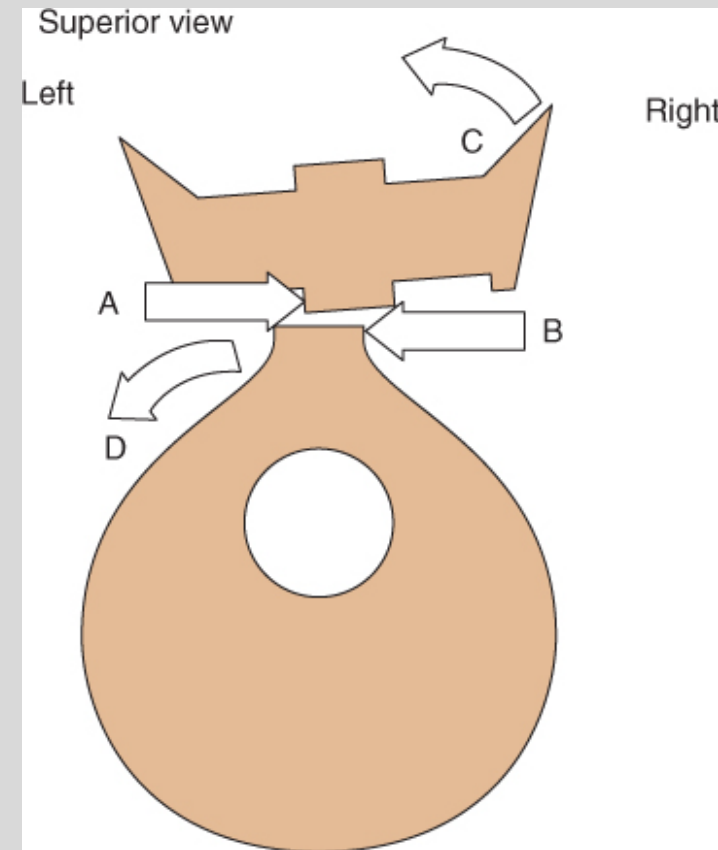




Lateral Strains



Left Lateral Strain
(feels like Greater Wing to
the Right)

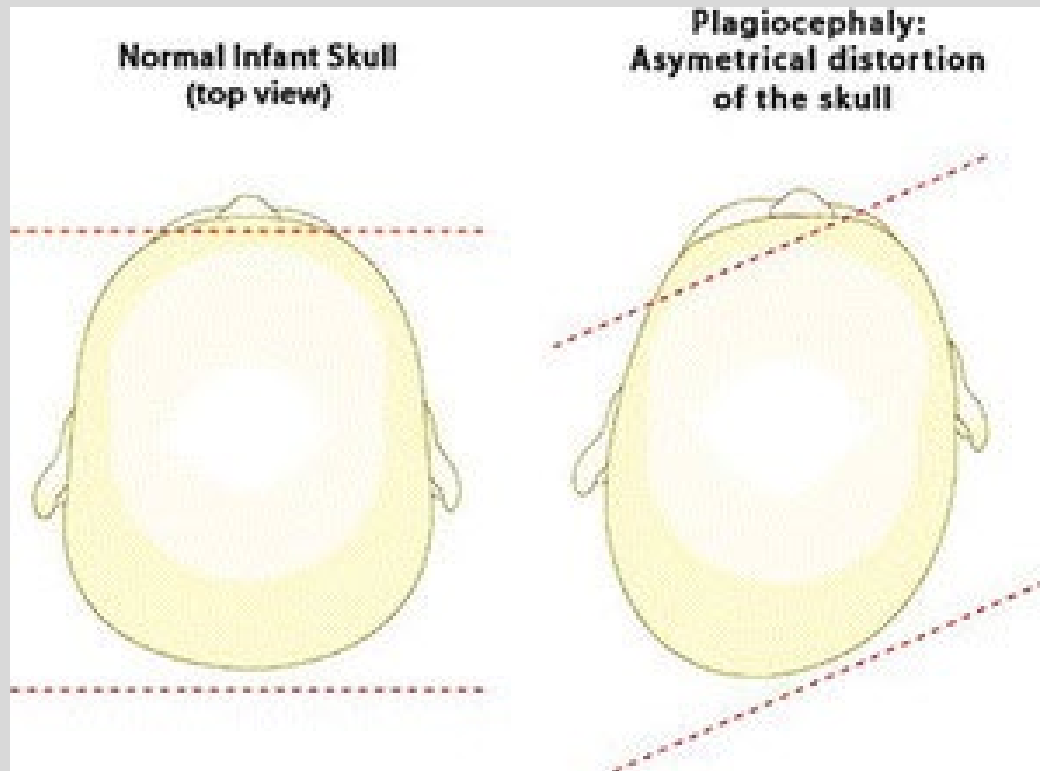


Right Lateral Strain
(feels like Greater Wing to
the Left)



Lateral Strains: “Parallelogram Head”

- Can be due to compressive force, positional, in utero or plagiocephaly secondary to torticollis or trauma from a lateral force



Lateral Strains

Left Lateral
strain GW to
the right



Neutral



Right Lateral
strain GW to
the left



- Index fingers move to side the greater wings (GW), the pinkies move to the opposite side (generally the direction of the basi-sphenoid)
- Hands will form a parallelogram.

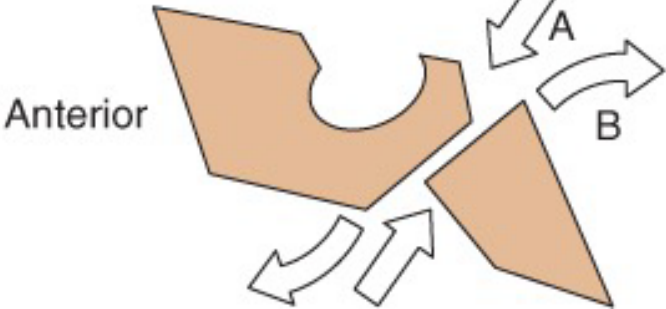
GW = Greater wing of the sphenoid

Case #3

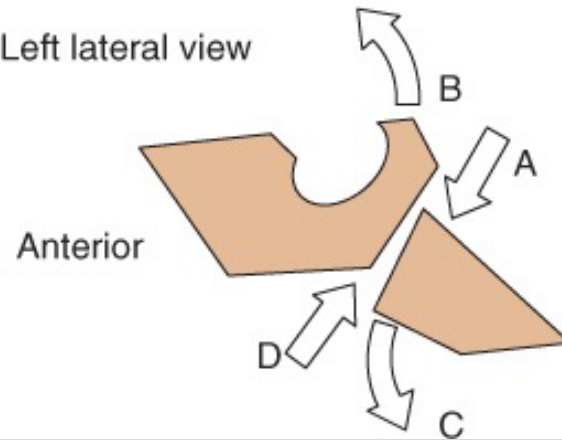
- A 50-year-old man presents with c/o dizziness, and a persistent ringing in his ear after tripping and hitting his chin on the dining room table as he fell to the ground.
- What cranial nerves are involved?
- What foramen does it exit from and what cranial bone is most involved?
- What strain pattern could have occurred in this case?

Putting it together - Vertical strains

Left lateral view



Left lateral view



Inferior vertical strain



Superior vertical strain



- Vertical Strains (Vertical Shears) – rotate same direction about their transverse axes
- Inferior Vertical Shear – both hands will rotate posteriorly, as if fingers are rotating up toward the ceiling
- Superior Vertical Shear- both hands will rotate anteriorly; or feels as if fingers are rotating down towards the table

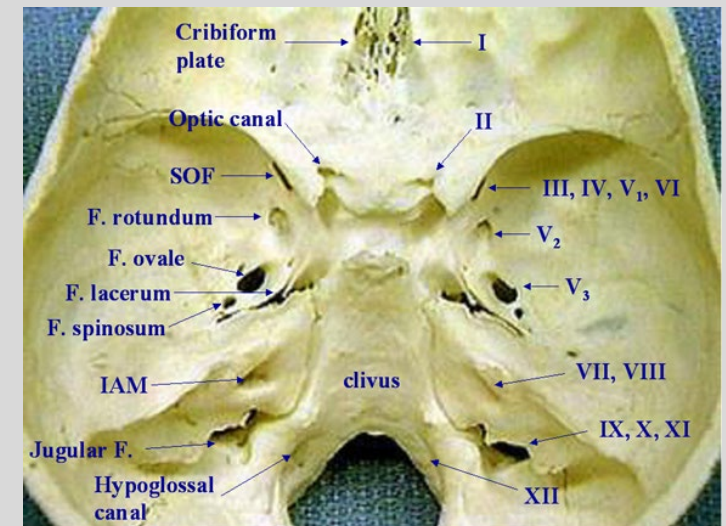
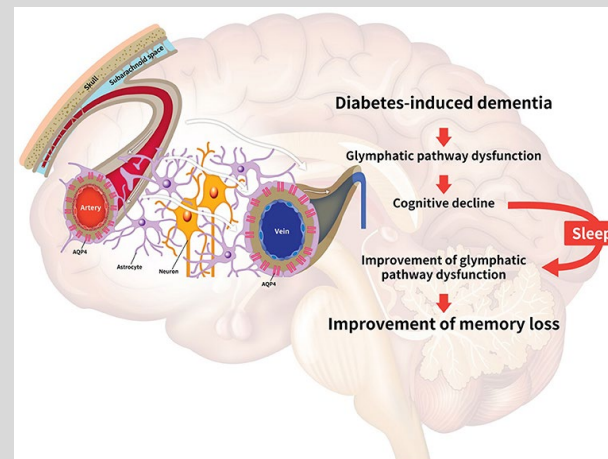
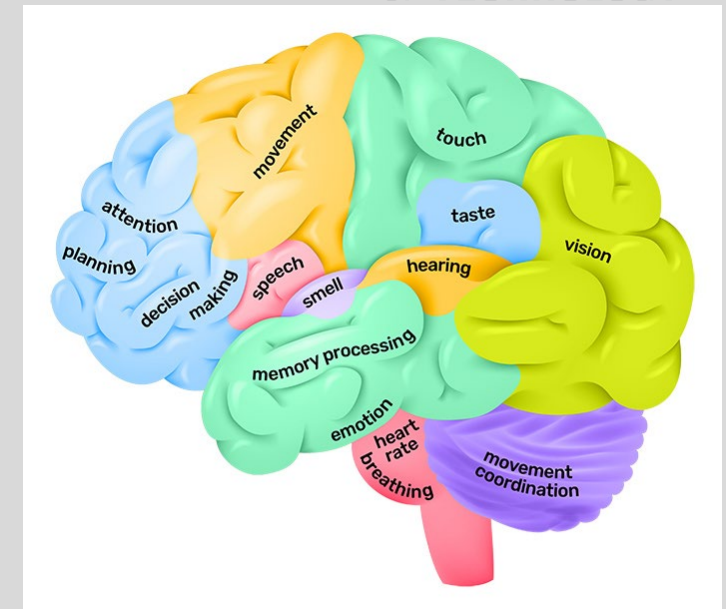
Case #3 continued

- You palpate your patient's head and don't feel any motion present. Other than doubting yourself and thinking you are a horrible osteopath what else could be the reason why you don't feel anything?



Clinical Significance

- Diagnosis and treatment of cranial strains can help to restore natural motion.
- Reduce potential cranial nerve compression and balance ANS.
- Enhance circulation and blood flow.
- Improve glymphatic drainage.
- Improve sinus drainage.



Osteopathy in the Cranial Field (OCF) – Cranial Nerves

- Olfactory nerve – CN I (Cribiform plate – ethmoid)
- Optic nerve – CN II (optic canal – sphenoid)
- Oculomotor nerve – CN III (superior orbital fissure (SOF) – sphenoid)
- Trochlear nerve – CN IV (SOF – sphenoid)
- Trigeminal nerve – CN V (V1 – SOF, V2 – Foramen Rotundum, V3 – Foramen Ovale – sphenoid)
- Abducens – CN VI (SOF – sphenoid & petrosphenoid ligament)

Osteopathy in the Cranial Field (OCF) – Cranial Nerves

- Facial nerve - CN VII (internal acoustic meatus – Temporal bone)
- Vestibulocochlear nerve – CN VIII (internal acoustic meatus – Temporal bone)
- Glossopharyngeal - CN IX (Jugular foramen - OM suture)
- Vagus – CN X (Jugular foramen - OM suture)
- Spinal Accessory – CN XI (Jugular foramen - OM suture)
- Hypoglossal – CN XII (Foramen magnum - OA & occipital intraosseous strain)

What Problems Can Potentially Be Treated with Cranial Osteopathic Manipulative Medicine (COMM)?

Dental

- * Orthodontic Problems
- * TMJ Syndrome
- * Malocclusion

Digestive

- * Constipation
- * Irritable Bowel Syndrome
- * Nausea

Ear, Nose, and Throat

- * Chronic Ear Infections
- * Chronic Tonsillitis
- * Sinusitis

Genitourinary

- * Bedwetting
- * Prostatitis
- * Stress Incontinence

Neurological

- * Dizziness and Vertigo
- * Head Trauma
- * Migraine Headaches
- * Post Concussion Syndrome
- * Post-MVA Inertial Injury (Whiplash)
- * Seizures
- * Visual Disturbances

Pediatric

- * ADD, ADHD
- * Autism
- * Behavioral Problems
- * Cerebral Palsy
- * Colic
- * Developmental Delays
- * Failure to Thrive
- * Learning Disorders
- * Otitis Media (ear infections)
- * Plagiocephaly (misshaped head)
- * Spitting Up
- * Strabismus (Crossed-Eyes or Lazy Eye)
- * Sucking Difficulties

- Osteopathy in the cranial field (OCF) can be utilized for many different conditions.
- Understanding the motions and patterns that can occur at the SBS can help with diagnosing and treating your patients.
- Utilizing your understanding of the anatomy and cranial nerves paths will guide your treatment with COMM.
- Applying COMM can help support patient health and healing from injuries.